SHIELD-AI™

A Sovereign Australian Framework for Secure Agentic AI

Security Architecture Practice • SHIELD-AI v1.11 • September 2026

THE AGENTIC AI SECURITY CHALLENGE

Australian enterprises are deploying AI agents that can **act autonomously** -- not just chatbots, but systems that:

78%

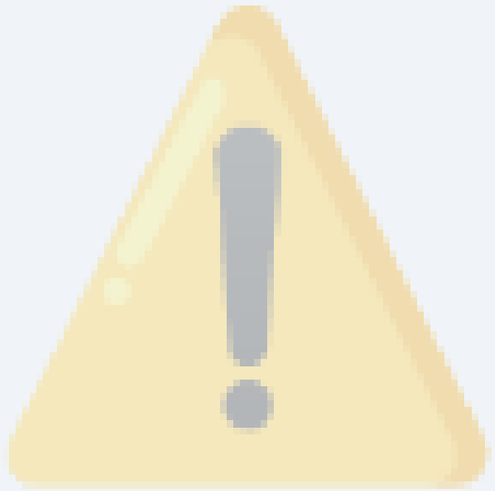
of AU enterprises deploying or piloting AI agents

63%

say existing security frameworks

12h

SOCI Act reporting window -- too fast for



"WE DIDN'T KNOW THE AGENT WAS COMPROMISED UNTIL THE REGULATOR

WHY EXISTING FRAMEWORKS AREN'T ENOUGH

Framework	Agentic AI Gap
ISM (1,081 controls)	No AI-specific controls
Essential Eight	Designed for commodity malware
NIST CSF / ISO 27001	No coverage of prompt injection, tool delegation, agent identity

THE RESULT

Organisations face a **compliance gap**: they meet their ISM/E8 obligations on paper but have no defensible position for agentic AI risk.

AUSTRALIAN REGULATORY IMPERATIVE

Your obligations don't stop at traditional infosec -- every Australian framework now intersects with AI:

ASD ISM (1,081 CONTROLS)

"Guidelines for secure AI system development" -- March 2026 edition

APRA CPS 234

Board accountability for information security -- including AI

SHIELD-AI FRAMEWORK

Seven pillars covering the full agentic AI lifecycle

PILLAR 1

MODEL SECURITY

Model weights, supply chain, adversarial robustness

PILLAR 2

IDENTITY & ACCESS

Agent identity, scope-bound delegation, human oversight

PILLAR 3

DATA SECURITY

Classification, minimisation, memory sanitisation

PILLAR 4

TOOL & PIPELINE

Tool registry, API gateway, pipeline integrity

PILLAR 5

MONITORING & IR

Behavioural detection, mandatory reporting, forensics

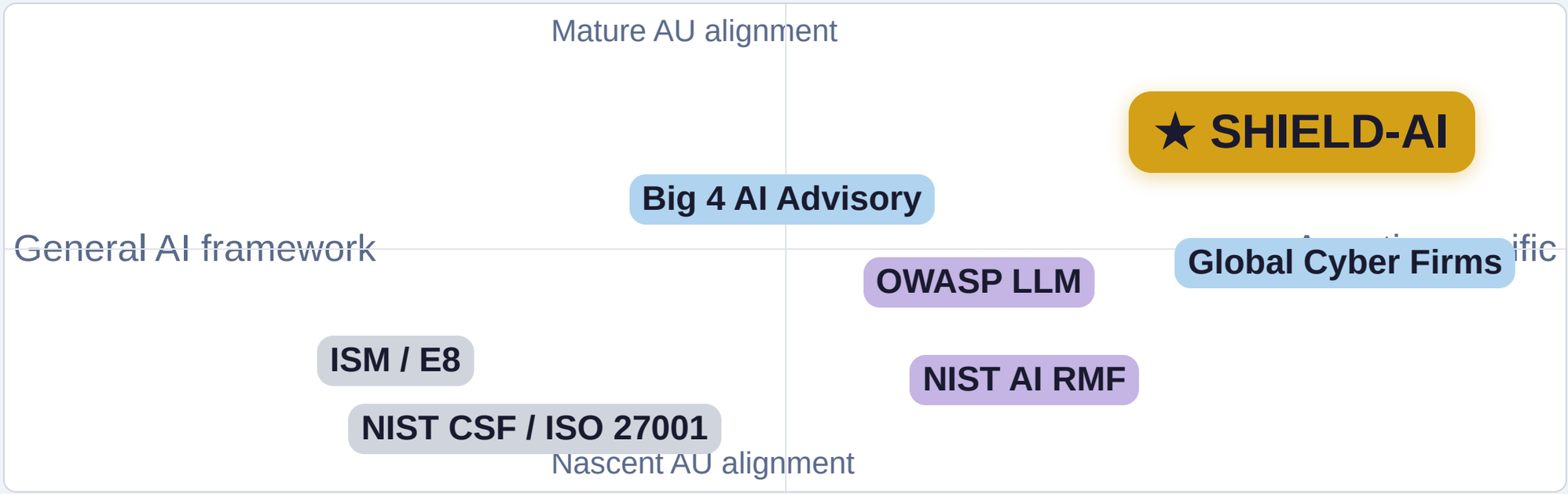
PILLAR 6

GOVERNANCE & RISK

AI risk register, board reporting, compliance

WHY SHIELD-AI?

Positioned to dominate -- agentic-specific controls with mature AU regulatory alignment



★ **SHIELD-AI** -- Only framework in the top-right: agentic-specific + AU mature. Everyone else covers only one dimension.

YOUR SHIELD-AI JOURNEY

Four maturity levels -- no organisation starts at L4

Level	Name	Description	Typical Client
L1	Initial	Basic controls, no formal governance	AI experimenters, PoCs
L2	Managed	Documented controls, basic monitoring	SMEs, early adopters
L3	Defined	Automated controls, regulatory mapped	Regulated entities, government
L4	Optimised	Continuous improvement, red-teaming	Critical infrastructure, national security

THE BUSINESS CASE

The cost of an unsecured agentic AI estate

Risk Scenario	Cost of Inaction
AI exfiltrates customer PII	$1.78M avg NDB breach cost (OAIC 2025) + regulatory action + reputation damage
Prompt injection causes financial loss	Loss amount x agent transaction limit + brand damage
SOCI-reportable incident	Penalties up to $2.2M per obligation + public notification
APRA regulator inquiry	Enforcement action + remediation costs
Board liability for AI	Director duty of care + civil penalties

MATURITY LEVELS

Four maturity levels -- adopt proportional to your risk profile

L1 INITIAL

Ad-hoc AI security; basic controls for model deployment

No formal governance

Shared credentials

L2 MANAGED

Documented controls for priority AI systems

Per-agent service accounts

Basic

L3 DEFINED

Enterprise-wide AI security program

Automated controls

Integrated monitoring

L4 OPTIMISED

Continuous improvement

AI-specific threat intelligence

Automated red-teaming

Proactive

HOW WE DELIVER

1 2 3 4 5

Discover	Assess	Design	Implement	Assure
Weeks 1-2 Inventory & regulatory map	Weeks 3-4 Gap analysis & maturity score	Weeks 5-8 Target architecture & roadmap	Weeks 9-16 Controls, policies, integration	Ongoing Reviews, monitoring, updates

WHAT A SHIELD-AI IMPLEMENTATION DELIVERS

AT L2 (MANAGED)

You know what agentic AI risks you have, which regulatory obligations apply, and what to fix first.

AT L3 (DEFINED)

Agentic AI at Maturity

🏅 **Major Australian Bank**

"Zero AI controls to L3 in 14 weeks. Regulator asked for AI security posture -- we sent the SHIELD-AI compliance dashboard."

🏳️ **Federal Government Agency**

"The initial assessment found 60% of AI pilots had no access controls. Implementation closed

NEXT STEPS

1. INITIAL CONSULTATION (30 MINUTES)

We assess your current AI posture, regulatory obligations, and readiness for a SHIELD-AI engagement.

2. ASSESSMENT SCOPING

If appropriate, we scope a SHIELD-AI assessment tailored to your environment and regulatory context.

SHIELD-AI — IN 90 SECONDS

SHIELD-AI is Australia's sovereign security framework for organisations deploying autonomous AI agents — systems that act, not just chat.

The problem: Existing frameworks (ISM, E8, NIST) don't cover AI-specific threats like prompt injection, tool delegation abuse, or autonomous decision-making. Yet ASIC, APRA, OAIC, and the government's proposed mandatory AI guardrails all expect regulated entities to demonstrate AI security.